On Construction of Approximate Real Mutually Unbiased Bases for an infinite class of dimensions $d \not\equiv 0 \bmod 4$

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Abstract

It is known that real Mutually Unbiased Bases (MUBs) do not exist for any dimension $d > 2$ which is not divisible by 4. Thus, the next combinatorial question is how one can construct Approximate Real MUBs (ARMUBs) in dimensions which are not divisible by 4. In this paper, for the first time, we show that it is possible to construct $\geq \lceil \sqrt{d} \rceil$ many ARMUBs for certain odd dimensions d of the form $d = (4n-t)s$, $t = 1, 2, 3$, where n is a natural number and s is an odd prime power. Considering the objects from combinatorial designs, our method exploits any available $4n \times 4n$ real Hadamard matrix H_{4n} (if exists) and uses this to construct an orthogonal matrix Y_{4n-t} of size $(4n-t) \times (4n-t)$, such that the absolute value of each entry varies a little from $\frac{1}{\sqrt{4n-t}}$. In our construction, the absolute value of the inner product between any pair of basis vectors from two different ARMUBs will be $\leq \frac{1}{\sqrt{d}}(1 + O(d^{-\frac{1}{4}})) < \frac{2}{\sqrt{d}}$, for proper choices of parameters, the class of dimensions d being infinitely large. Finally we connect this with Welch Bounds related to spherical codes.

Key Words: Combinatorial Design, Hadamard Matrices, Mutually Unbiased Bases, Quantum Information Theory, Resolvable Block Design.

1 Introduction

Mutually Unbiased Bases (defined in definition 1) are important combinatorial objects that received serious attention in Quantum Information Processing. Such structures are useful in different aspects of Quantum Cryptology

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and Communications. We refer to [9, 10, 11] towards detailed motivations regarding this domain of research.

It is well known that given any finite dimensional Hilbert space $\mathbb{C}^d$, the number of MUBs can at most be $d+1$. In spite of considerable efforts for several decades, the cases reaching the upper bound could only be constructed when d is a prime power. For a specific dimension d, other than the prime powers, constructing a larger number of MUBs (upper bounded by $d+1$) is believed to be one of the most challenging problems in Quantum Information Theory. The situation becomes further restrictive when we consider the elements of the MUBs to be real. A pair of MUBs can always be transformed into a pair of an identity matrix and a Hadamard matrix, by doing a unitary transformation from the left, as a unitary transformation preserves the inner products.

For dimensions, that are not prime powers, constructing large number of MUBs remains elusive. At this point, various kinds of Approximate MUBs have been explored. There are results in this domain where such approximate structures have been constructed using various algebraic structures [3, 14, 16, 17, 19, 21], combinatorial designs [9, 10, 11] and computational search methods [4].

When MUBs are constructed over $\mathbb{R}^d$, one can obtain real MUBs. In most dimensions, a large number of real MUBs are non-existent [2]. In fact, only for $d = 4^s, s > 1$, one can obtain $\frac{d}{2}+1$ many real MUBs which is maximum for any dimension d, whereas for most of the dimensions d, which are not perfect square, we have at best only 2 real MUBs [2]. One may refer to [2, Table 1] for a comprehensive view regarding this. This provides the motivation to construct Approximate Real MUBs (ARMUBs) as evident in recent literature [9, 10, 21].

Given $d = p_1^{n_1} p_2^{n_2} \ldots p_s^{n_s}$, where each p_i is a distinct prime and each n_i is a positive integer, the lower bound on the number of MUBs in $\mathbb{C}^d$ is $p_r^{n_r}+1$, where $p_r^{n_r} = \min\{p_1^{n_1}, p_2^{n_2}, \ldots, p_s^{n_s}\}$. Naturally, constructing a good number of MUBs for any composite dimension proved to be almost elusive even over $\mathbb{C}^d$. Naturally with additional restrictions, the number of such bases is very small [2] when we consider the problem over the real vector space $\mathbb{R}^d$. This is summarized in [15, Theorem 3.2] too.

In this regard, one must note that there cannot be any pair of real MUBs for dimensions which are > 2 and not divisible by 4. This is immediately understood as there cannot be any real Hadamard matrix in such dimensions and existence of a pair of real MUBs provides a Hadamard matrix. With this motivation, in this paper, we consider construction of Approximate Real MUBs (ARMUBs) for dimensions $d \not\equiv 0 \bmod 4$ where $d > 2$. For a broad description of various technical terms and contexts related to this work, we refer to the papers [9, 10, 11].

1.1 Organization & Contribution

This paper is organized as follows. In Section 2, we define MUBs and the approximate variants. We outline the existing ideas with examples that will be used in our contributions. In Section 3, we construct the orthogonal matrices of orders $4n-t,\ t=1,2,3$, from real Hadamard matrix H_{4n} of order $4n$. We name these as ϵ-Hadamard matrices. Finally, using such matrices and suitable Resolvable Block Designs (RBDs), we construct ARMUBs for dimensions $d=(4n-t)q$, for $t=1,2,3$. We will show that when q is an odd prime power, and the order of $4n$ and q are the same, then we can have order of $\sqrt{d}$ many ARMUBs with significantly low inner product values.

To compare with the earlier works, we will refer to [9, 10, 11].

- In [9], the ARMUBs are constructed for the dimensions $d=(4x)^2$, where x is a prime power. The number of such ARMUBs was $\frac{\sqrt{d}}{4}+1$, and the value of the inner products was $\leq \frac{4}{\sqrt{d}}$.
- This has been improved in [10], where ARMUBs for dimensions $d=q(q+1)$, when prime power $q \equiv 3 \bmod 4$ were considered. The inner products were improved to $< \frac{2}{\sqrt{d}}$, and $\lceil \sqrt{d} \rceil$ many such ARMUBs could be obtained. In certain cases results were available for $d=sq^2$, where q is a prime power and $sq \equiv 0 \bmod 4$.
- Later APMUBs (Almost Perfect) MUBs were studied in [11], and several constructions over complex numbers were considered. The constructions could be achieved for reals, but in those cases the dimensions are again divisible by 4, as they require existence of real Hadamard matrices.

In this paper[1] (Subsection 3.1) we modify the real Hadamard matrices to obtain the real unitary ones of order $(4n-t)$, for $t=1,2,3$, which we refer as ϵ-Hadamard. These are used in conjunction with RBDs in line of the constructions in [10, 11] to obtain new constructions of β-ARMUBs for the dimensions $d>2$ and $d \not\equiv 0 \bmod 4$, with $\beta<2$. In algebraic terms, this bound considering the inner products between two vectors from two different ARMUBs is $\leq \frac{1}{\sqrt{d}}(1+O(d^{-\frac{1}{4}}))$. When s is an odd prime power, and the order of $4n-t$ and s are same, we obtain $O(\sqrt{d})$ many ARMUBs for such dimensions. Note that even a pair of real MUBs are not available for the dimensions $d>2$ and $d \not\equiv 0 \bmod 4$ and thus the approximate versions are the only relevant solutions in this direction, that we first present in this paper. We also present a coding theoretic application in Subsection 3.4 related to Welch Bounds [20].

[1] A few proofs are omitted in this version and those are available in [12].

2 Preliminaries & Background

Let us first formally introduce the MUBs.

Definition 1. By $\mathbb{C}^d$, let us denote a d-dimensional complex vector space. Then consider two orthonormal bases, $M_l = \left\{ |\psi_1^l\rangle, |\psi_2^l\rangle, \ldots, |\psi_d^l\rangle \right\}$ and $M_m = \{ |\psi_1^m\rangle, |\psi_2^m\rangle, \ldots, |\psi_d^m\rangle \}$, in $\mathbb{C}^d$. Two such bases will be called Mutually Unbiased if we have $\left| \langle \psi_i^l | \psi_j^m \rangle \right| = \dfrac{1}{\sqrt{d}}$, $\forall i, j \in \{1, 2, \ldots, d\}$. The set $\mathbb{M} = \{M_1, M_2, \ldots, M_r\}$ consisting of such orthonormal bases will form MUBs of size r, if every pair in the set is mutually unbiased.

When the two bases are not mutually unbiased, but almost close to that, then we call them Approximate MUBs (AMUBs) [9, 10].

Definition 2. The two bases will be called Approximately Mutually Unbiased if we have

$$\left| \langle \psi_i^l | \psi_j^m \rangle \right| \approx \frac{1}{\sqrt{d}}, \ \forall i, j \in \{1, 2, \ldots, d\}. \tag{1}$$

The set $\mathbb{M} = \{M_1, M_2, \ldots, M_r\}$ consisting of such orthonormal bases will form Approximate MUBs (AMUBs) of size r, if every pair in the set is approximately mutually unbiased. This approximation needs to be formalized and it is required that the deviation from $\frac{1}{\sqrt{d}}$ should be as small as possible. When the bases are real, we call them Approximate Real MUBs (ARMUBs).

In this regard, there are two more definitions.

Definition 3. A set of r orthonormal bases M_i, $0 \leq i \leq r-1$ of $\mathbb{R}^d$ are called β-ARMUBs if for any two bases M_l and M_m, we have

$$\left| \langle \psi_i^l | \psi_j^m \rangle \right| \leq \frac{\beta}{\sqrt{d}}, \ \forall i, j \in \{1, 2, \ldots, d\}. \tag{2}$$

Definition 4. [11] The set $\mathbb{M} = \{M_1, M_2, \ldots, M_r\}$ will be called Almost Perfect MUBs (APMUBs) if $\Delta = \left\{ 0, \frac{\beta}{\sqrt{d}} \right\}$, i.e., the set contains just two values, such that $\beta = 1 + O(d^{-\lambda}) \leq 2$, for any real $\lambda > 0$. When the bases are real, we call them Almost Perfect Real MUBs (APRMUBs). In this context we denote Δ for the set containing different values of $\left| \langle \psi_i^l | \psi_j^m \rangle \right|$ for $l \neq m$.

While studying approximate mutually unbiased bases (AMUBs) over $\mathbb{C}^d$, several constructions have been proposed in [3, 14, 17]. However, these approaches are not applicable for constructing real mutually unbiased bases. To address this, [9] introduced a construction for Approximate Real MUBs (ARMUBs) using real Hadamard matrices. Specifically, the work in [9] showed that for dimensions of the form $d = (4q)^2$, where q is a prime, it is possible to construct $\frac{\sqrt{d}}{4}+1$ ARMUBs such that the maximum inner product between vectors from distinct bases is $\frac{4}{\sqrt{d}}$.

This result had subsequently been generalized and improved in [10] by exploiting RBDs. In particular, the techniques in [10] extended the construction to dimensions $d = sq^2$, where q is a prime power and s is some integer, and improved the achievable parameters. It was demonstrated in [10] that for $d = q(q+1)$, q being a prime power satisfying $q \equiv 3 \pmod 4$, one can construct $\lceil \sqrt{d} \rceil = q + 1$ ARMUBs such that the maximum inner product between vectors belonging to different bases is upper bounded by $\frac{2}{\sqrt{d}}$. Thus, the improvement in [10] is two-fold.

- Firstly, it yields a larger number of bases.
- Secondly, it provides a smaller upper bound on the inner product values compared to [9].

Further advancements in this direction, particularly for Approximate Partial Real MUBs (APRMUBs), have been reported in [11], although that work did not address constructions in odd dimensions.

It is also worth noting that in the complex space $\mathbb{C}^d$, each Mutually Unbiased Basis (MUB) comprises d orthonormal vectors, which can be represented as the columns of a $d \times d$ unitary matrix. A collection of MUBs is therefore equivalent to a set of unitary matrices. Without loss of generality, one of these matrices can be mapped to the identity matrix I via a unitary transformation. For instance, if we have r many MUBs $\{M_1, M_2, M_3, \ldots, M_r\}$ in $\mathbb{C}^d$, where $r \leq d+1$, each represented as a unitary matrix, then left-multiplying each matrix by M_1^{-1} yields the transformed set $\{I, M_1^{-1}M_2, M_1^{-1}M_3, \ldots, M_1^{-1}M_r\}$. Since the inverse of a unitary matrix is its conjugate transpose, calculating $M_i^{-1}M_j$ for $i \neq j$ involves taking inner products between the columns of the matrices. As a result, the modulus of each entry in the resulting product matrix is $\frac{1}{\sqrt{d}}$. Taking $\frac{1}{\sqrt{d}}$ as a common factor, the remaining matrix has entries of unit modulus, and thus forms a complex Hadamard matrix.

Some basic examples related to real MUBs for dimension 2, 3, 4 and the basis of our construction are explained in details in [12].

3 Our Construction

In this section we present our technique to have ARMUBs, which were not presented earlier in literature. For construction of ARMUBs, using [12, Construction 1] (see also [10, 11]), we make use of RBD(X, A), with $|X| = d = k \times s$, where $s > k$ and s, k are as close as possible, so that both s, k are of $O(\sqrt{d})$. Since $d > 2$ is not multiple of 4, thus k or s cannot be a multiple of 4. For constructing ARMUBs corresponding to each parallel class of an RBD, we exploit suitable orthogonal matrix Y, which we call (as in Definition 5 later) ϵ-Hadamard Matrix of order k, modifying a real Hadamard one of order $k+t$ (where $k+t$ is divisible by 4) such that $|(Y)_{ij}|$, the absolute value of each entry $(Y)_{ij}$ is very close to $\frac{1}{\sqrt{k}}$. The Hadamard conjecture tells that in such cases we have the existence, and thus we can consider $t = \{1, 2, 3\}$ for our purpose. Let us now consider how do we modify the Hadamard matrices.

3.1 Modifying real Hadamard matrices

As discussed, we will construct orthogonal matrix of order $k \in \{4n-1, 4n-2, 4n-3\}$ from Hadamard matrix of order $4n$, using the method presented below. We will call such orthogonal matrices as ϵ-Hadamard Matrices. We define the notion of ϵ-Hadamard matrices of order k, which in some sense can be considered as close counterpart of Hadamard in such order, where no Hadamard matrix exists ($k \neq 4n$) and will be helpful in constructing β-ARMUBs with good properties. The basic property of Hadamard matrix which is helpful in such construction, is the same absolute value of all the entries, which is $\frac{1}{\sqrt{4n}}$. Taking cue from this, we intend to construct orthogonal matrix for order $k \neq 4n$, such that the absolute value of the entries of the matrix are very close to each other. In this regard, let us present the following definition for our purpose.

Definition 5. For an orthogonal matrix $\mathbb{O}_{k\times k}$, if the absolute value of each entry of this matrix $|(\mathbb{O}_{k\times k})_{ij}| = \frac{1+\epsilon_{ij}}{\sqrt{k}}$, where $\epsilon_{ij} = \pm O(k^{-\lambda}) < 1$, such that $\lambda > 0$, then we call the orthogonal matrix $\mathbb{O}_{k\times k}$ an ϵ-Hadamard matrix, when $\epsilon = \max_{i,j} |\epsilon_{ij}|$.

If there are choices of many such matrices, then we will consider the one with least ϵ available at hand. When k is a multiple of 4, assuming Hadamard conjecture, it is obvious that the minimal value of ϵ is zero, but when $k > 2$ is not a multiple of 4, finding the global minima of ϵ appears to be a very challenging problem. However, we first show in Theorem 1 that such result with $\epsilon < 1$ can be achieved. Later, in Section 3.2, we will consider various U and correspondingly, we will obtain the orthogonal matrix with the least ϵ.

We now have the following technical result. Please note that with certain abuse of notations, we write $U_{t\times t}$ to express that it is a $t \times t$ matrix, but while explaining we only write U as we can refer the (i, j)-th element as subscript such as U_{ij} or $(U)_{ij}$. We let $\mathbb{I}_{t\times t}$ refer to the identity matrix of order $t \times t$.

Lemma 1. Let $N = \frac{1}{\sqrt{m}} \begin{bmatrix} U_{t\times t} & V_{t\times(m-t)} \\ W_{(m-t)\times t} & D_{(m-t)\times(m-t)} \end{bmatrix}_{m\times m}$ be an $m \times m$ orthogonal matrix, containing block matrices U, V, W, and D of sizes as indicated. If either $(\mathbb{I} + \frac{1}{\sqrt{m}}U)$ or $(\mathbb{I} - \frac{1}{\sqrt{m}}U)$ is a nonsingular matrix, then $Y_1 =$

$\frac{1}{\sqrt{m}}\left(D - \frac{1}{\sqrt{m}}W(\mathbb{I} + \frac{1}{\sqrt{m}}U)^{-1}V\right)$ and $Y_2 = \frac{1}{\sqrt{m}}\left(D + \frac{1}{\sqrt{m}}W(\mathbb{I} - \frac{1}{\sqrt{m}}U)^{-1}V\right)$ are both orthogonal matrices of order $(m-t)$.

For a matrix of the form $(\mathbb{I} \pm \frac{1}{\sqrt{m}}U)$, which is relevant for demonstrating specific constructions, we have following result.

Lemma 2. Let $\kappa, \gamma, \alpha, \vartheta$ be real.

1. If $U^2 = \kappa\mathbb{I} + \gamma U$, then the inverse of the matrix $\left(\mathbb{I} + \frac{1}{\alpha}U\right)$ exists if $\alpha^2 + \gamma\alpha - \kappa \neq 0$ and is given by $\frac{\alpha(\alpha+\gamma)}{\alpha^2+\gamma\alpha-\kappa}\left(\mathbb{I} - \frac{1}{\alpha+\gamma}U\right)$.

2. If $U^3 = \kappa\mathbb{I} + \gamma U + \vartheta U^2$, then the inverse of the matrix $\left(\mathbb{I} + \frac{1}{\alpha}U\right)$ exists if $\alpha^3 + \vartheta\alpha^2 - \gamma\alpha + \kappa \neq 0$ and is given by
$$\frac{\alpha(\alpha(\alpha+\vartheta)-\gamma)}{\alpha^3+\vartheta\alpha^2-\gamma\alpha+\kappa}\left(\mathbb{I} - \frac{\alpha+\vartheta}{\alpha(\alpha+\vartheta)-\gamma}U + \frac{1}{\alpha(\alpha+\vartheta)-\gamma}U^2\right).$$

In the above lemma, if U satisfies the item 1, then we should use that, and one may note that the solution from item 2 will be the same. In case, item 1 is not satisfied, then we should exploit item 2. This is because, the computation in the second case will be more involved.

Note that when U is 2×2 matrix, we will have $U^2 = \gamma U + \kappa\mathbb{I}$, where $\gamma = \mathrm{Tr}(U)$ and $\kappa = -\mathrm{Det}(U)$. When U is a 3×3 matrix, $U^3 = \vartheta U^2 + \gamma U + \kappa\mathbb{I}$ where $\vartheta = \mathrm{Tr}(U)$, $\gamma = \sum_{i>j}(U)_{ij}(U)_{ji} - (U)_{ii}(U)_{jj}$ and $\kappa = \mathrm{Det}(U)$.

Let us now concentrate on eigen value characterization. When $U^2 = \kappa\mathbb{I} + \gamma U$ then the eigen values of U are the roots of the equation $\lambda^2 - \gamma\lambda - \kappa = 0$, and hence has maximum two different values, say $\{\lambda_1, \lambda_2\}$. Thus, U is similar to diagonal matrix $\mathrm{Diag}(\lambda_1, \ldots, \lambda_1, \lambda_2, \ldots \lambda_2)$. Similarly when $U^3 = \kappa\mathbb{I} + \gamma U + \vartheta U^2$ then eigen values of U are the roots of the equation $\lambda^3 - \vartheta\lambda^2 - \gamma\lambda - \kappa = 0$ and in this case U has maximum three different eigen values, say $\{\lambda_1, \lambda_2, \lambda_3\}$ and hence U is similar to diagonal matrix $\mathrm{Diag}(\lambda_1, \ldots \lambda_1, \lambda_2, \ldots \lambda_2, \lambda_3, \ldots \lambda_3)$.

We use this and apply the above Lemma 1 for the case when N is a real Hadamard matrix of order $m = 4n$, denoting it by H_{4n} and obtain the following result.

Theorem 1. Let $H_{4n} = \frac{1}{\sqrt{4n}}\begin{bmatrix} U_{t\times t} & V_{t\times(4n-t)} \\ W_{(4n-t)\times t} & D_{(4n-t)\times(4n-t)} \end{bmatrix}_{n\times n}$ containing block matrices U, V, W, and D of sizes as indicated. If $t < \sqrt{n}$ then $Y_1 = \frac{1}{\sqrt{4n}}\left(D - \frac{1}{\sqrt{4n}}W(\mathbb{I} + \frac{1}{\sqrt{4n}}U)^{-1}V\right)$, $Y_2 = \frac{1}{\sqrt{4n}}\left(D + \frac{1}{\sqrt{4n}}W(\mathbb{I} - \frac{1}{\sqrt{4n}}U)^{-1}V\right)$ are ϵ-Hadamard matrices of order $(4n-t)$ with $\epsilon = \frac{t}{\sqrt{k}} + \mathcal{O}(k^{-1}) \leq 1$, where
$$\frac{1}{\sqrt{4n}}\left(1 - \frac{t}{\sqrt{4n}-t}\right) \leq |(Y_{1,2})_{ij}| \leq \frac{1}{\sqrt{4n}}\left(1 + \frac{t}{\sqrt{4n}-t}\right).$$

As there are various options for real Hadamard matrices, when they exist, for $t \geq 1$, there are various possibilities of U too and corresponding to each possibility, we will have different Y_1, Y_2. Thus, we should consider which one can be chosen for the best result among the various possibilities of U, we are considering.

3.2 Explaining the cases with $t = 1, 2, 3$

We present explicit constructions of orthogonal matrices $Y_{(4n-t)\times(4n-t)}$ (i.e., Y_1 or Y_2 as in Theorem 1) by considering the block matrix $U_{t\times t}$. The matrix U denotes a $t \times t$ matrix whose entries are all ± 1. For $t = 1$, we have $U = \pm 1$. Given one element, without loss of generality, fix $U_1 = 1$. For $t = 2$, as there are 4 elements, we have $2^4 = 16$ possible configurations of U. For $t = 3$, there are $2^9 = 512$ possible configurations.

We have examined several (not all) cases out of these and here we present the results for the ϵ-Hadamard matrix to have the least ϵ among the ones we considered below. As we have pointed out, H_{4n} be an orthogonal matrix of the form

$$H_{4n} = \frac{1}{\sqrt{4n}} \begin{bmatrix} U_{t\times t} & V_{t\times(4n-t)} \\ W_{(4n-t)\times t} & D_{(4n-t)\times(4n-t)} \end{bmatrix},$$

where $t < \sqrt{4n}$ and all submatrices are real. In particular, we need to consider the cases for $t = 1, 2, 3$ and obtain the following result.

Theorem 2. For $t = 1, 2, 3$, it is possible to construct real ϵ-Hadamard matrices of dimension $4n - t$ with $\epsilon = \frac{\rho_t}{\sqrt{n}}$, where $\rho_t = \frac{1}{2}, 2$ and 4 respectively for $t = 1, 2, 3$.

Programs in Python are implemented in this regard to experiment with such ϵ-Hadamard matrices, and the repository is available at [6]. We have noted that the actual values of ϵ are even smaller than the expressions in Theorem 2.

3.3 Construction of ARMUBs

As we have already discussed, construction for Approximate MUBs using Hadamard and other unitary matrices in conjunction with Resolvable Block Designs have been presented in [9, 10, 11]. The basic concept is to utilize an RBD(X, A), with $|X| = d = k \times s$, with a constant block size k, such that it has as many parallel classes as possible, but any pair of block from different parallel classes, should have maximum 1 point in common ($\mu = 1$). Here we need to use the unitary matrix of the order of block size k, to convert each parallel class into an orthonormal basis.

Given this context, we present the main result as follows.

Theorem 3. Let $d = k \times s$, such that $s > k$ and s is a power of odd prime. Let ϵ-Hadamard matrix Y of order k exist. Then one can obtain $s \geq \lceil \sqrt{d} \rceil$ many β-ARMUBs, with $\beta \leq 1 + O(d^{-\frac{1}{4}})$, which is < 2 for an infinite class of dimensions d.

Proof. We broadly consider [12, Construction 1], to obtain the orthonormal basis vectors in $\mathbb{R}^d$ corresponding to each parallel class in $\mathrm{RBD}(X, A)$ using ϵ-Hadamard matrix Y. Following [10, 11] the absolute value of the dot product of any pair of basis vectors u, v from different orthonormal basis $| \langle u, v \rangle | \leq \mu |(Y)_{ij}|^2_{\max} \leq (\frac{1+\epsilon}{\sqrt{k}})^2 = \frac{(1+\epsilon)^2}{k}$, where μ is the intersection number and in this case any pair of blocks from different parallel classes will have maximum one point in common, i.e., $\mu = 1$. When $s - k > 0$ and s a power of prime, then one can construct $\mathrm{RBD}(X, A)$ having s many parallel classes, each with s many blocks of constant block size k (refer to [11, Construction 3, Lemma 5] for exact details).

With proper manipulations of constants and noting the power of primes are at least as dense as the primes, we will obtain an odd prime power $s > \sqrt{d}$, in an expected distance $O(\log d)$ from $\sqrt{d}$. Thus, considering Theorem 2 for the values of ϵ, we obtain the inner product value between two vectors from two different ARMUBs $\leq \frac{(1+\epsilon)^2}{k} \leq \frac{1}{\sqrt{d}}(1 + O(d^{-\frac{1}{4}}))$. This gives $\beta \leq 1 + O(d^{-\frac{1}{4}})$. □

Example 1. Let us now consider an example for $d = 79 \times 3^4$, where $k = 79, s = 3^4$. In this case there are $79 + 1 = 80$ complex MUBs, but no pair of real ones. The existing works in this regard [9, 10, 11] do not consider this case too. In this effort, we will obtain $3^4 = 81$ many ARMUBs, which is $\lceil \sqrt{d} \rceil + 1$. As $79 = 4 \times 20 - 1$, from Theorem 2, we have $\epsilon < \frac{1}{2\sqrt{n}} = \frac{1}{2\sqrt{20}}$. Now, $k = 4n - 1 = 79$, thus, the numeric expression will be $\beta < 1.252$.

In case, we consider $k = 43$, instead of 79, we obtain $\beta \leq 1.697$. Naturally, we obtain the value of β closest to 1 (from above), when k, s are very close.

3.4 Applications to coding theory

Viewed as points on a hyper-sphere, the basis vectors of ARMUBs form spherical codes with $O(d^{\frac{3}{2}})$ codewords, which can be shown to approach the Welch Bounds [20] for large d, implying an asymptotically optimal packing. Further the ARMUBs constructed here are also very sparse, which makes the codewords formed useful in Vector Quantization, Compressing sensing, CDMA and other practical applications in signal processing.

It is clear that each ARMUB has d many vectors and there are $O(\sqrt{d})$ such ARMUBs, and hence we have a total of $O(d^{\frac{3}{2}})$ codewords in a spherical code. For any set of N unit vectors in $\mathbb{R}^d$, the maximum inner product magnitude (coherence τ) is bounded below, called the Welch Bounds, given by $\tau_{max} \geq \sqrt{\frac{N-d}{d(N-1)}}$. That is, the Welch Bounds state that for any set of N unit vectors in $\mathbb{R}^d$, the maximum inner product τ_{max} satisfies $\tau_{Welch} = \sqrt{\frac{N-d}{d(N-1)}}$ for $N \leq d^2$.

The result from Theorem 3 shows that, we have N as $O(d^{\frac{3}{2}})$, and thus, $\tau_{Welch} \approx \frac{1}{\sqrt{d}}$ when N is $O(d^{\frac{3}{2}})$, i.e., $N \leq d^2$. We have also proved that $\beta \leq 1+O(d^{-\frac{1}{4}})$, that is asymptotically $\beta \to 1$, and hence our result achieves the inner product value $\frac{1}{\sqrt{d}}$ asymptotically, reaching the Welch Bounds.

4 Conclusion

In this work, we presented constructions of β-ARMUBs for certain dimensions d that are not multiples of 4, when d is of the form $(4n-t)s$, where s is an odd prime power. Specifically, we obtain $\beta = 1+O(d^{-\frac{1}{4}}) < 2$, which is the upper bound of the inner product between any two vectors from two different MUBs. The number of ARMUBs available for these cases are $\geq \lceil \sqrt{d} \rceil$. Such classes of ARMUBs, for dimensions not divisible by 4, was not presented earlier. Further, the results have implications to coding theory.

However, our constructions do not yield real APMUBs (Almost Perfect Real MUBs or APRMUBs) since the set Δ in our case cannot guarantee $|\Delta| = 2$, which is mandatory for an APMUB construction, i.e., in our case, there may be more than 2 values of inner products. As no pair of real MUBs exists for dimensions $d \not\equiv 0 \bmod 4$, such constructions for APRMUBs would be an interesting research direction.

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